Chapter 23 (Carey Chapter 22)

The Chemistry of Amines

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Chapter 23 Overview

23.1 – Nomenclature of Amines

23.2 – Structure of Amines

23.3 – Physical Properties of Amines

23.4 – Spectroscopy of Amines

23.5 – Basicity and Acidity of Amines

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23.7 – Alkylation and Acylation Reactions of Amines

23.8 – Hofmann Elimination of Quaternary Ammonium Hydroxides

23.9 – Aromatic Substitution Reactions of Aniline Derivatives

23.10 – Diazotization; Reactions of Diazonium Ions

23.11 – Synthesis of Amines

23.12 – Use and Occurrence of Amines

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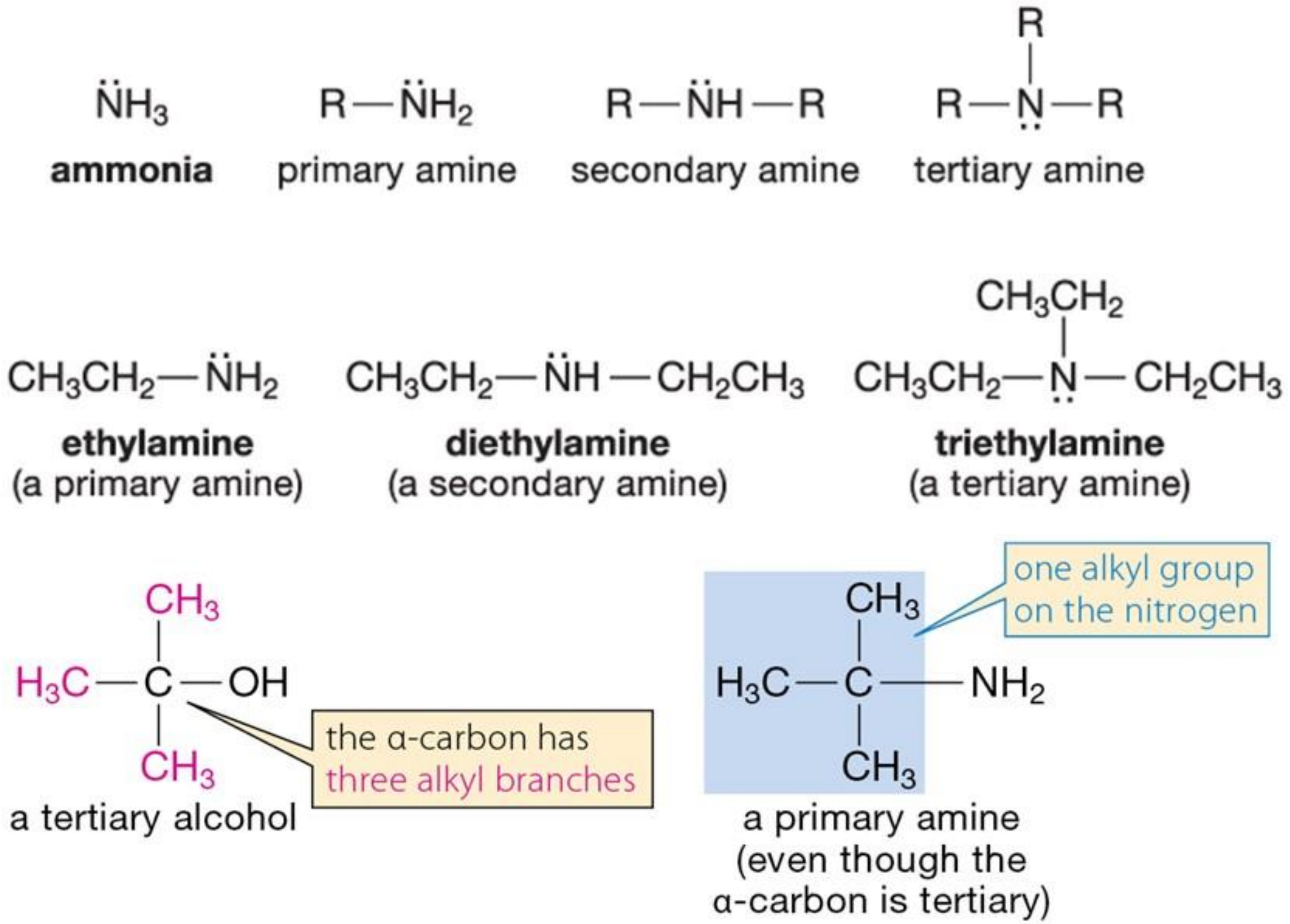
Classification of Amines

- Amines are organic derivatives of ammonia.
- Note the difference between the classification of amines and alcohols.



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Common Nomenclature

- Append *amine* to the name of the alkyl group.

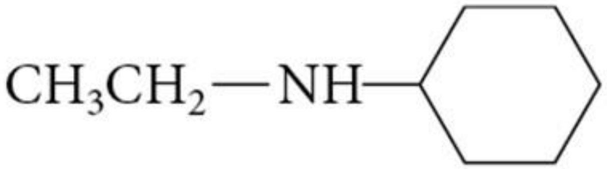
$CH_3CH_2NH_2$ **ethylamine**

$(CH_3)_3N$ **trimethylamine**

- For two or more different groups it is named as an *N*-substituted derivative of the larger group.

$(CH_3)_2N—CH_2CH_2CH_2CH_3$

***N,N*-dimethylbutylamine**



***N*-ethylcyclohexylamine**

- More information about amine nomenclature available at the end of this power point

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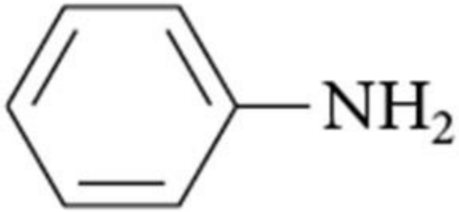
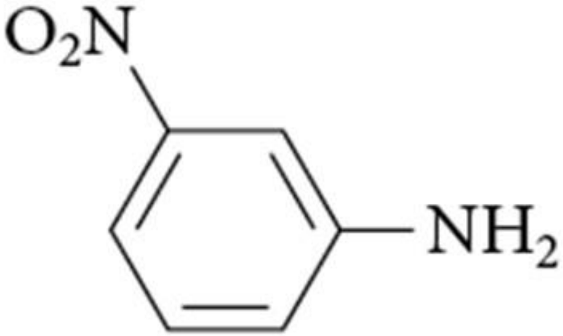
Common Nomenclature: Aromatic Amines

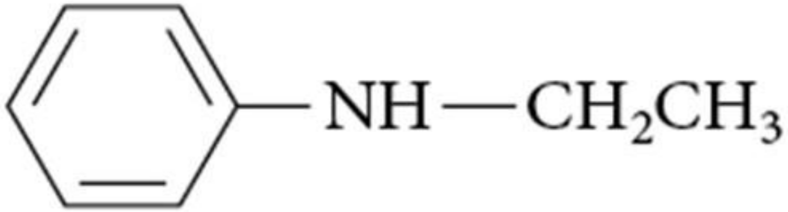
- Aromatic amines are named as derivatives of aniline:



aniline



3-nitroaniline
(*m*-nitroaniline)



***N*-ethylaniline**

- More information about amine nomenclature available at the end of this power point

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Basicity of Amines

- Amines are strong enough bases that they are completely protonated in dilute acidic solution.

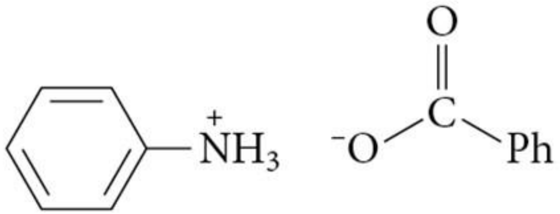
$$H_3C{-}\ddot{N}H_2 + H{-}Cl \rightleftharpoons H_3C{-}\overset{+}{N}H_3\ Cl^- + H_2O$$

methylamine | **methylammonium chloride**

- The salts of protonated amines are called ***ammonium salts.***

$$(CH_3)_2\overset{+}{N}H_2\ \ Cl^-$$

dimethylammonium chloride



anilinium benzoate
(anilin~~e~~ + *ium*)

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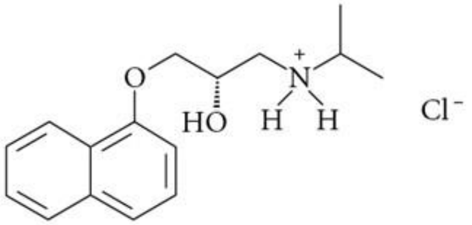
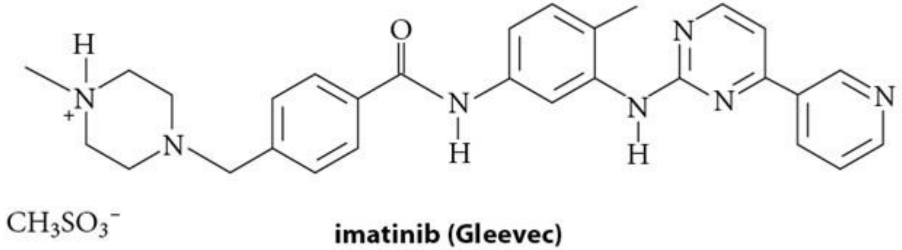
Amine Salts

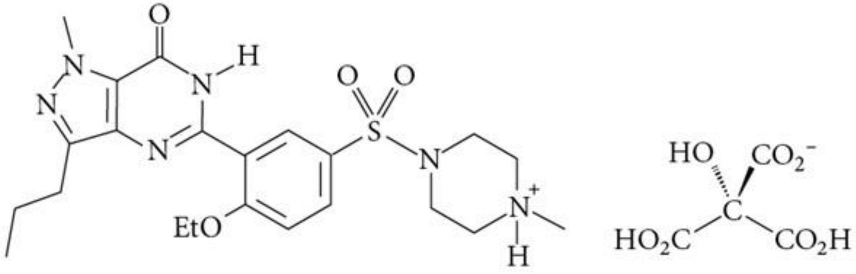
- Many pharmaceuticals containing amine functional groups are marketed as salts.



(*S*)-propranolol hydrochloride (Inderal)
(used to treat hypertension and anxiety)



imatinib (Gleevec)
(leukemia treatment)



sildenafil citrate (Viagra)
(a phosphodiesterase type 5 inhibitor)

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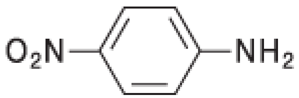
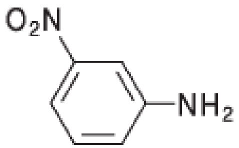
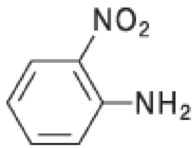
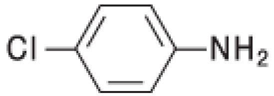
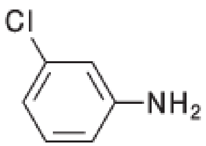
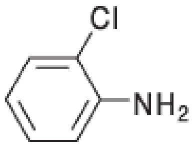
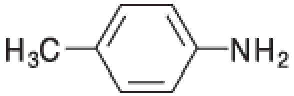
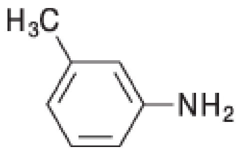
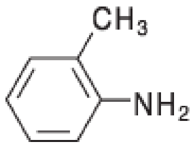
Basicities of Some Amines

Table 23.1 Basicities of Some Amines
(Each pK_a values is for the dissociation of the corresponding conjugate-acid ammonium ion.)

Amine	**pK_a**	**Amine**	**pK_a**	**Amine**	**pK_a**
CH_3NH_2	10.62	$(CH_3)_2NH$	10.64	$(CH_3)_3N$	9.76
$CH_3CH_2NH_2$	10.63	$(CH_3CH_2)_2NH$	10.98	$(CH_3CH_2)_3N$	10.65
$PhCH_2NH_2$	9.34				
$PhNH_2$	4.62	$PhNHCH_3$	4.85	$PhN(CH_3)_2$	5.06
	≈1.0		2.45		−0.26
	3.81		3.32		2.62
	5.07		4.67		4.38

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Acidity of Amines

- NH_3, RNH_2, and R_2NH are ***amphoteric***: they may act as bases and acids.
- They are very weakly acidic (unstable conjugate base).
- The conjugate base of an amine is called an ***amide*** (do not confuse with amide derivatives of carboxylic acids).

$$2\,Na + 2\,\ddot{N}H_3 \xrightarrow{Fe^{3+}} 2\,(Na^+ \;\; {}^{-}\!:\!\ddot{N}H_2) + H_2$$

sodium amide
(sodamide)

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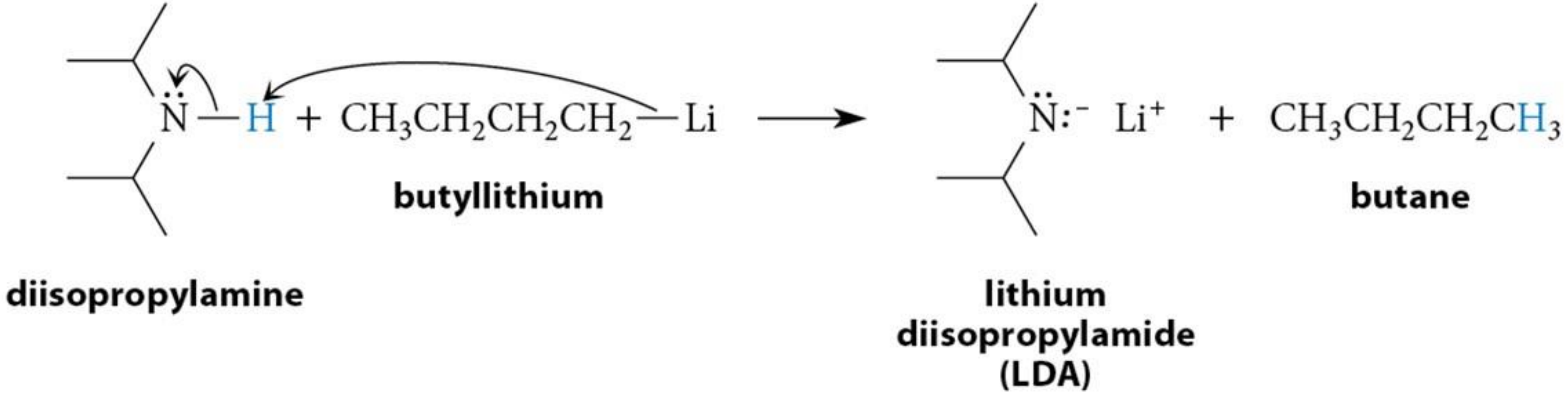
Preparation of Amides

- Amides are readily prepared by treating the amine with butyllithium (in THF or ether).



$R_2\ddot{N}—H$
amines
$pK_a \approx 32–35$

$R_3\overset{+}{N}—H$
ammonium ions
$pK_a \approx 9–11$

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Quaternary Ammonium Salts

- ***Quaternary ammonium salts*** are compounds in which all four groups around the nitrogen are alkyl or aryl.

$(CH_3)_4N^+\ Cl^-$

tetramethylammonium chloride

$Ph\overset{+}{N}Et_3\ Br^-$

***N,N,N*-triethylanilinium bromide**

$PhCH_2\overset{+}{N}(CH_3)_3\ {}^-OH$

benzyltrimethylammonium hydroxide
(Triton B)

$$CH_3(CH_2)_{15}-\overset{\overset{\displaystyle CH_3}{|}}{\underset{\underset{\displaystyle CH_3}{|}}{N^+}}-CH_2-Ph\quad Cl^-$$

"benzalkonium chloride"
a cationic surfactant (Sec. 20.5)

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Amines are Bronsted Bases; React as Nucleophiles

1. S_N2 reaction with alkyl halides, sulfonic esters or epoxides.
2. Addition to aldehydes, ketones and α,β-unsaturated carbonyl compounds.
3. Nucleophilic acyl substitution at the carbonyl carbons of carboxylic acid derivatives.

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Direct Alkylation of Amines with Halides

$$R_3N\!: + H_3C{-}I \longrightarrow R_3\overset{+}{N}{-}CH_3\ I^-$$

- Further alkylations can take place to give complex mixtures:

$$\ddot{N}H_3 + CH_3I \longrightarrow CH_3\overset{+}{N}H_3\ I^- + (CH_3)_2\overset{+}{N}H_2\ I^- + (CH_3)_3\overset{+}{N}H\ I^- + (CH_3)_4N^+\ I^-$$

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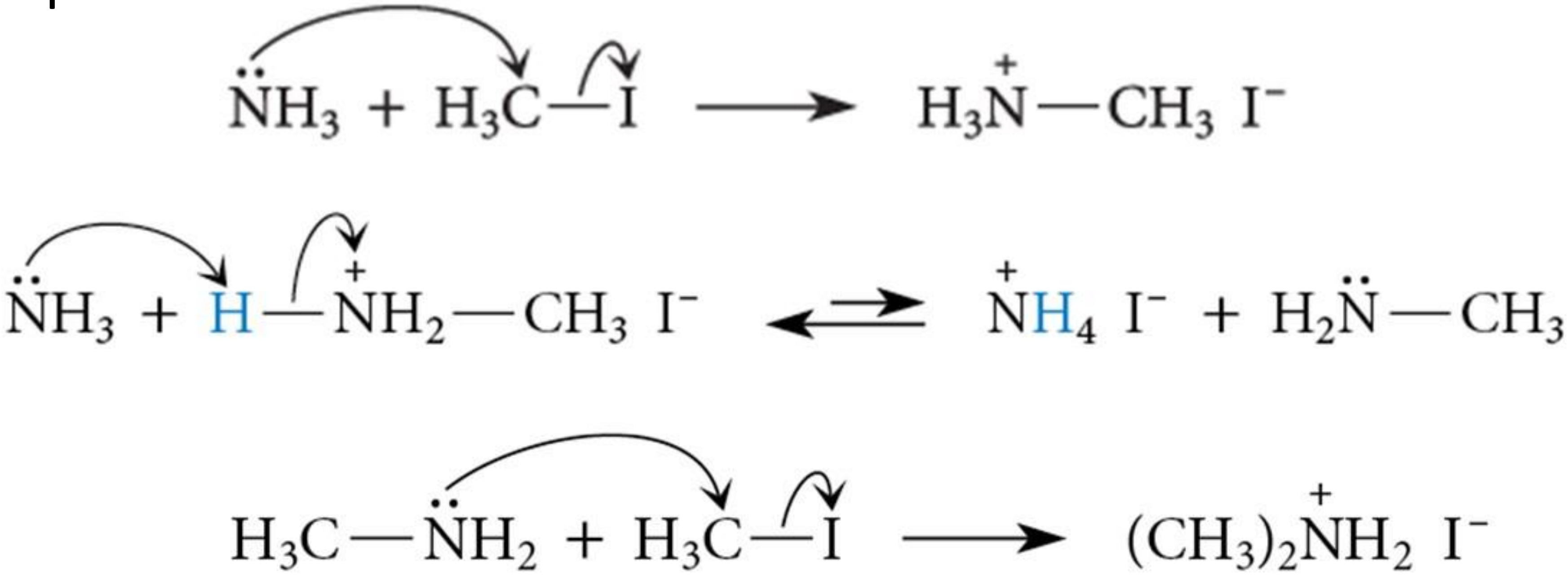
Mechanism Producing Complex Mixture

- Successive alkylation–deprotonation events account for the formation of these products.



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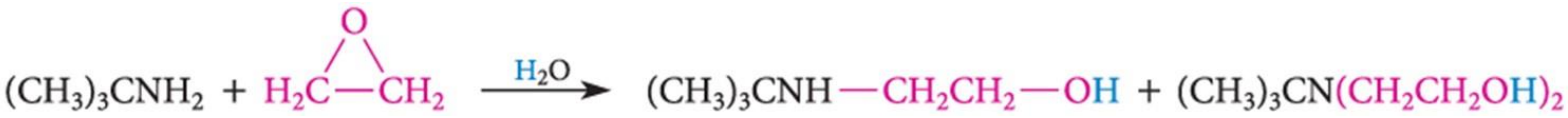
Alkylation of Amines: Epoxides and α,β-Unsaturated Carbonyls and Nitriles



$$NH_3\ \text{(excess)} + H_2C{=}CH{-}CN \longrightarrow \underset{\text{(32\% yield)}}{H_2N{-}CH_2CH_2CN} + \underset{\text{(57\% yield)}}{HN(CH_2CH_2CN)_2}$$

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Direct Alkylation of Amines: Quaternization

- Important synthetic application of amine alkylation

$$PhCH_2\ddot{N}Me_2 + MeI \xrightarrow{EtOH} PhCH_2\overset{+}{N}Me_3\ I^-$$

benzyldimethylamine → **benzyltrimethylammonium iodide** (94–99% yield)

$$CH_3(CH_2)_{15}NMe_2 + PhCH_2{-}Cl \xrightarrow{acetone} CH_3(CH_2)_{15}\overset{+}{N}(CH_2Ph)Me_2\ Cl^-$$

***N,N*-dimethyl-1-hexadecanamine** + **benzyl chloride** → **benzylhexadecyldimethylammonium chloride**

$$CH_3CH(CH_2CH_3)NHMe + MeI\ (excess) \xrightarrow[ether]{heat} CH_3CH(CH_2CH_3)\overset{+}{N}Me_3\ I^- + HI$$

***sec*-butylmethylamine** → ***sec*-butyltrimethylammonium iodide**

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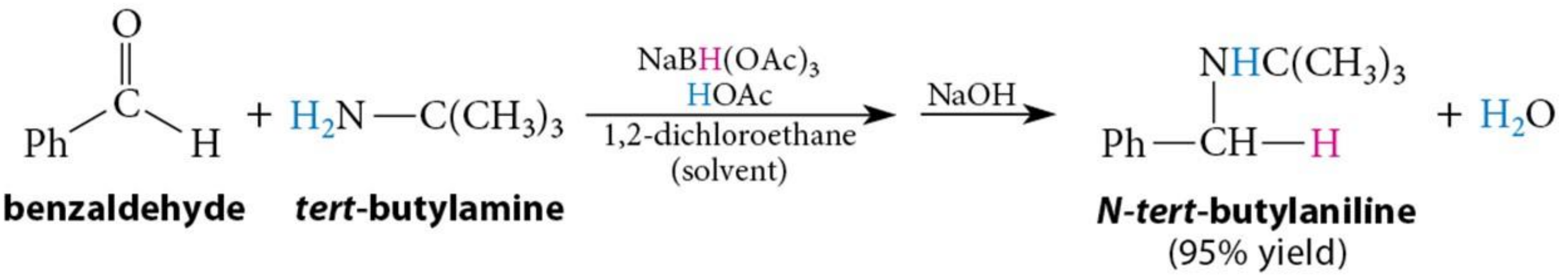
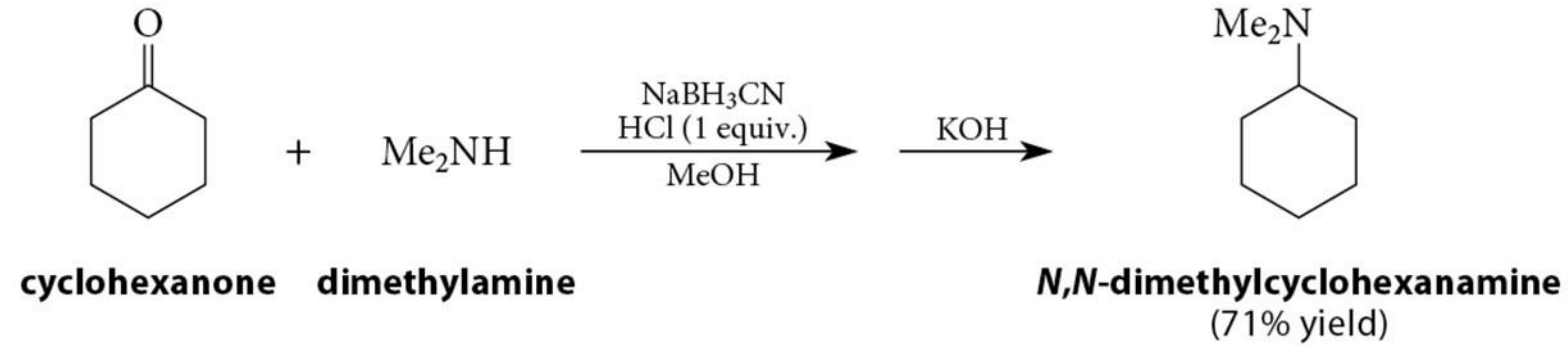
Reductive Amination: Hydride Reducing Agents





23.7 Alkylation and Acylation Reactions of Amines

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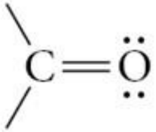
Recap: Reactions with Primary Amines

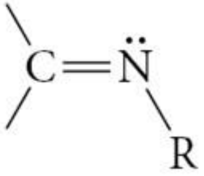
- Reaction of a primary amine with an aldehyde or ketone provides an imine product (also called a ***Schiff base*** or a ***Schiff's base***).

$R-\ddot{N}H_2$

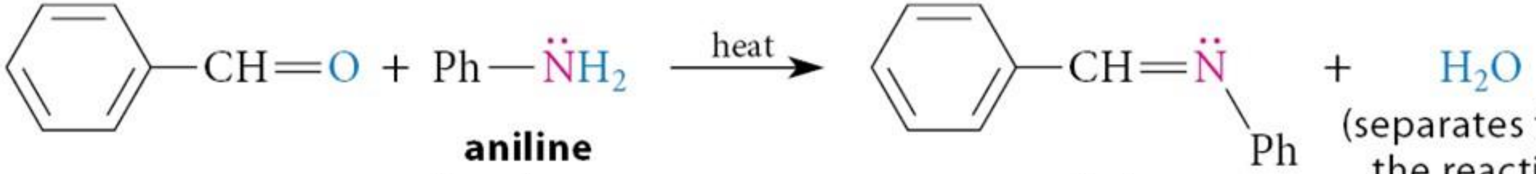
primary amine



aldehyde or ketone



imine
(Schiff base)



aniline
(a primary amine)

an imine
(84–87% yield)

(separates from the reaction mixture)

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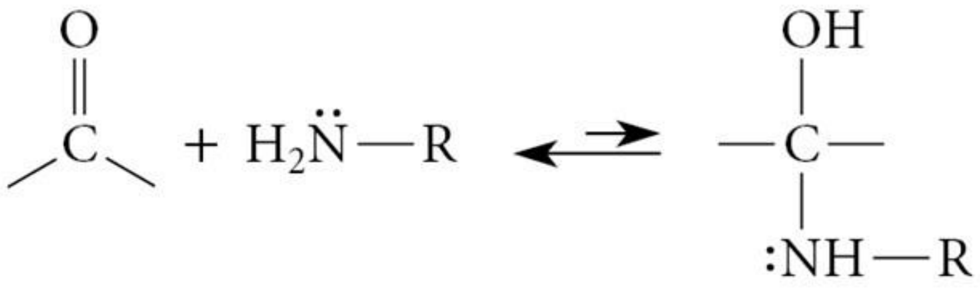
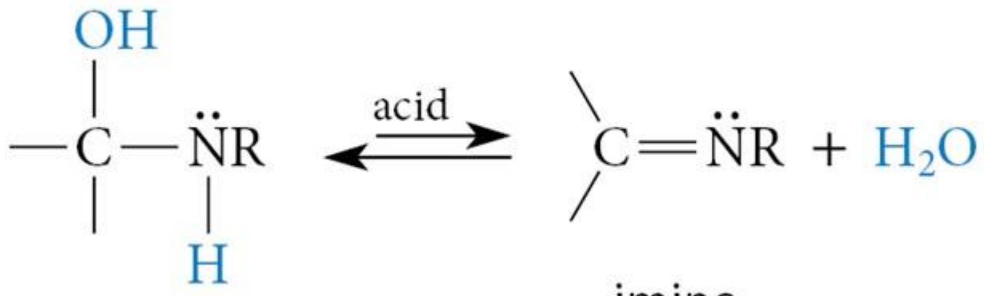
Recap: Mechanism of Reaction with Primary Amines

- Formation of carbinolamine
- The carbinolamine dehydration is typically the rate-limiting step.



carbinolamine



carbinolamine

imine
(Schiff base)

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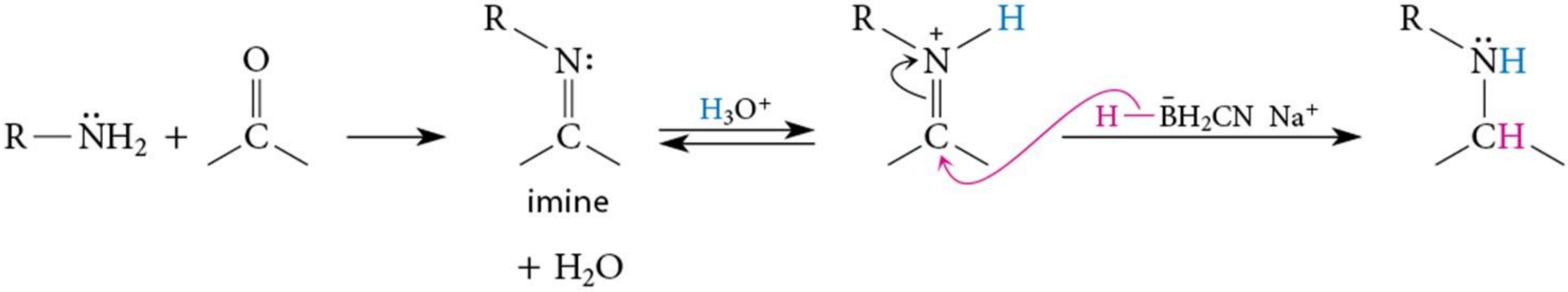
Reductive Amination: Imine Intermediate



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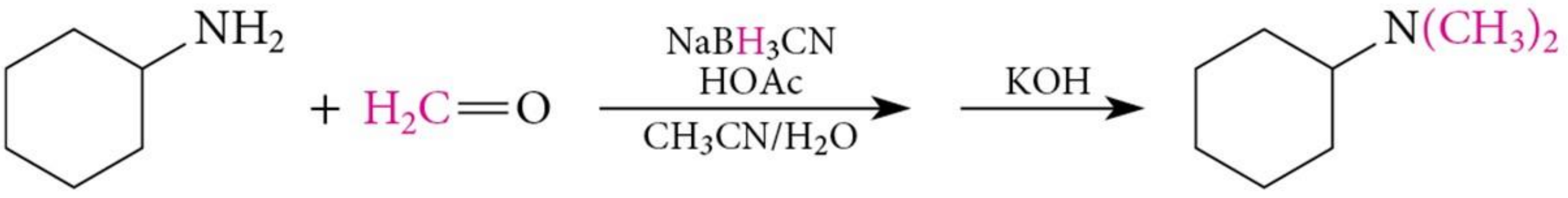
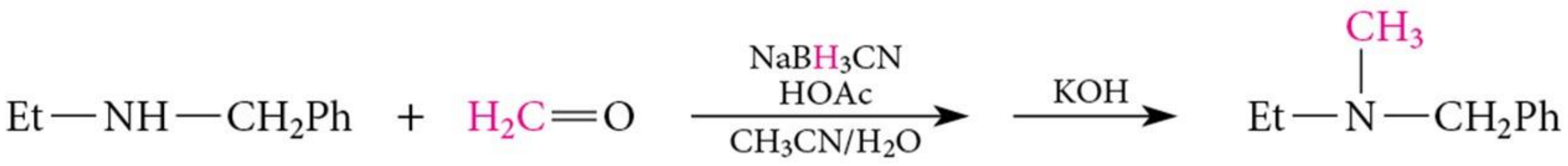
Reductive Amination: Introducing Methyl Groups

- Formaldehyde can be used to introduce methyl groups



(84% yield)



benzylethylamine **formaldehyde** **benzylethylmethylamine**
(80% yield)

- Comparison of borohydride reagents on the board.

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Following slides just for your information

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Substitutive Nomenclature (1 of 2)

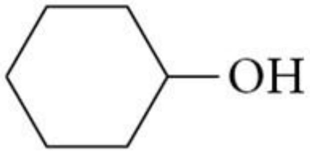
- The most widely used system of substitutive amine nomenclature is that of *Chemical Abstracts.*
- In this system, the suffix ***amine*** is used.

$$\begin{array}{c} OH \\ | \\ CH_3CHCH_2CH_2CH_3 \end{array}$$

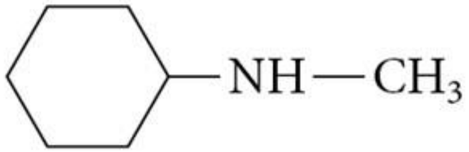
2-pentanol

$$\begin{array}{c} NH_2 \\ | \\ CH_3CHCH_2CH_2CH_3 \end{array}$$

2-pentanamine



cyclohexanol



***N*-methylcyclohexanamine**

$$H_2N—CH_2CH_2CH_2CH_2CH_2—NH_2$$

1,5-pentanediamine

$$H_3C—NH—CH_2CH_2CH_2—NH—CH_2CH_3$$

***N*-ethyl-*N′*-methyl-1,3-propanediamine**

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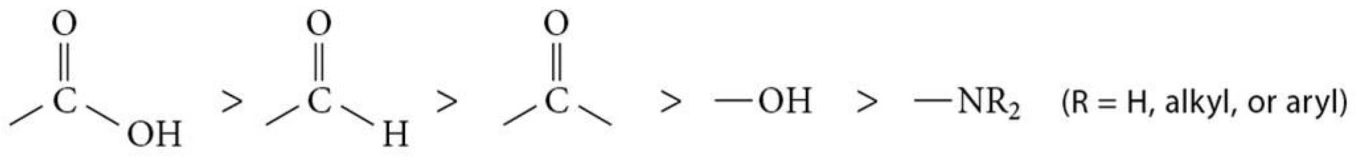
Substitutive Nomenclature (2 of 2)

- The priority for amines is as follows:



(carboxylic acid and derivatives) (aldehyde) (ketone) (alcohol) (amine)

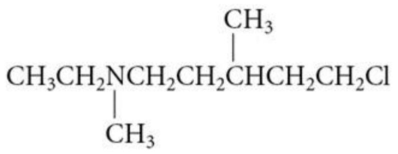
- When cited as a substituent, the $-NH_2$ group is called the ***amino*** group.

$H_2N—CH_2CH_2—OH$

2-aminoethanol
(OH has priority)

$H_2N—\overset{1}{C}H_2—\overset{2}{C}H{=}\overset{3}{C}H_2$

2-propen-1-amine
(NH_2 has priority)



5-chloro-*N*-ethyl-*N*,3-dimethyl-1-pentanamine

$(CH_3)_2N—CH_2CH_2CH_2—OH$

3-(dimethylamino)-1-propanol

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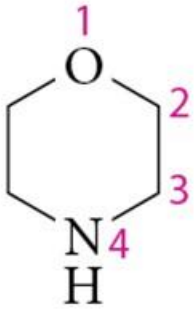
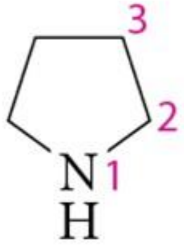
Some Important Heterocyclic Amines

- Numbering begins with the heteroatom.

piperidine

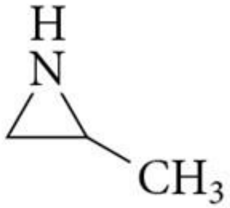


morpholine

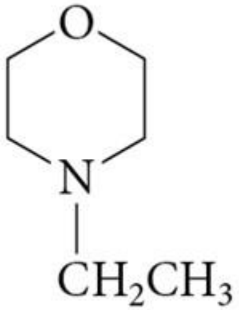


pyrrolidine

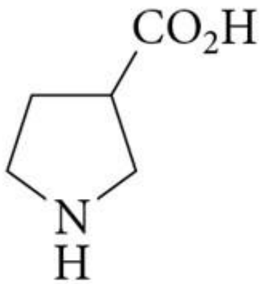
aziridine



2-methylaziridine



***N*-ethylmorpholine**
(4-ethylmorpholine)



3-pyrrolidinecarboxylic acid

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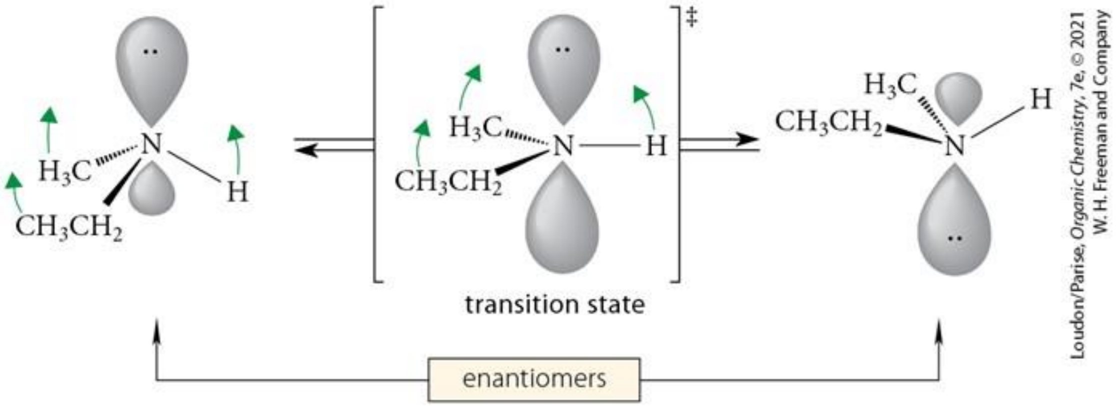
Structure of Amines

- Bond length:

bond:	C—C	C—N	C—O	C–F
typical length:	154 pm	147 pm	143 pm	139 pm

- Most amines undergo rapid inversion at the nitrogen (typically no chirality on the nitrogen):



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Physical Properties of Amines

- Most amines are liquids with unpleasant odors (fishy, putrid).
- H-bonding ability increases boiling point.

	$CH_3CH_2\overset{CH_3}{\overset{\vert}{C}}HCH_3$	$CH_3CH_2N(CH_3)_2$	$(CH_3CH_2)_2NH$	$CH_3CH_2CH_2CH_2NH_2$
	isopentane	***N,N*-dimethylethylamine**	**diethylamine**	**butylamine**
boiling point:	27.8 °C	37.5 °C	56.3 °C	77.8 °C
dipole moment:	0 D	0.6 D	1.2–1.3 D	1.4 D

Increasing hydrogen bonding and polarity →

	$(CH_3CH_2)_2NH$	$(CH_3CH_2)_2O$	$(CH_3CH_2)_2CH_2$	$CH_3CH_2CH_2CH_2OH$	$CH_3CH_2CH_2CH_2NH_2$
	diethylamine	**diethyl ether**	**pentane**	**1-butanol**	**1-butanamine**
boiling point:	56.3 °C	37.5 °C	36.1 °C	117.3 °C	77.8 °C

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Four Factors Influence the Basicity of Amines

1. The effect of alkyl substitution
2. The polar effect
3. The resonance effect
4. The effect of charge

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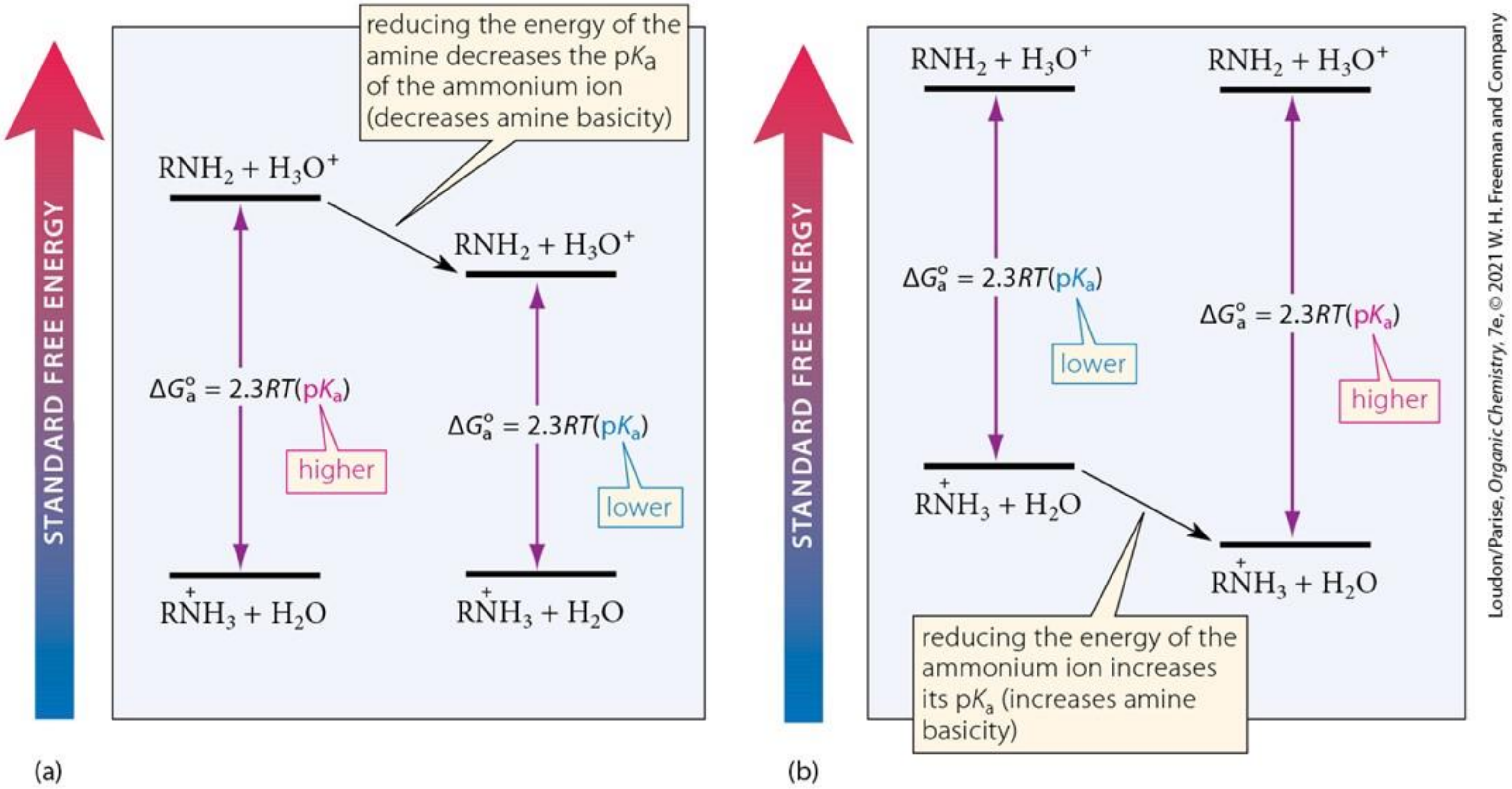
Substituent Effects on Amine Basicity

- If a substituent stabilizes the ammonium ion more than its conjugate base amine, the amine basicity is increased.



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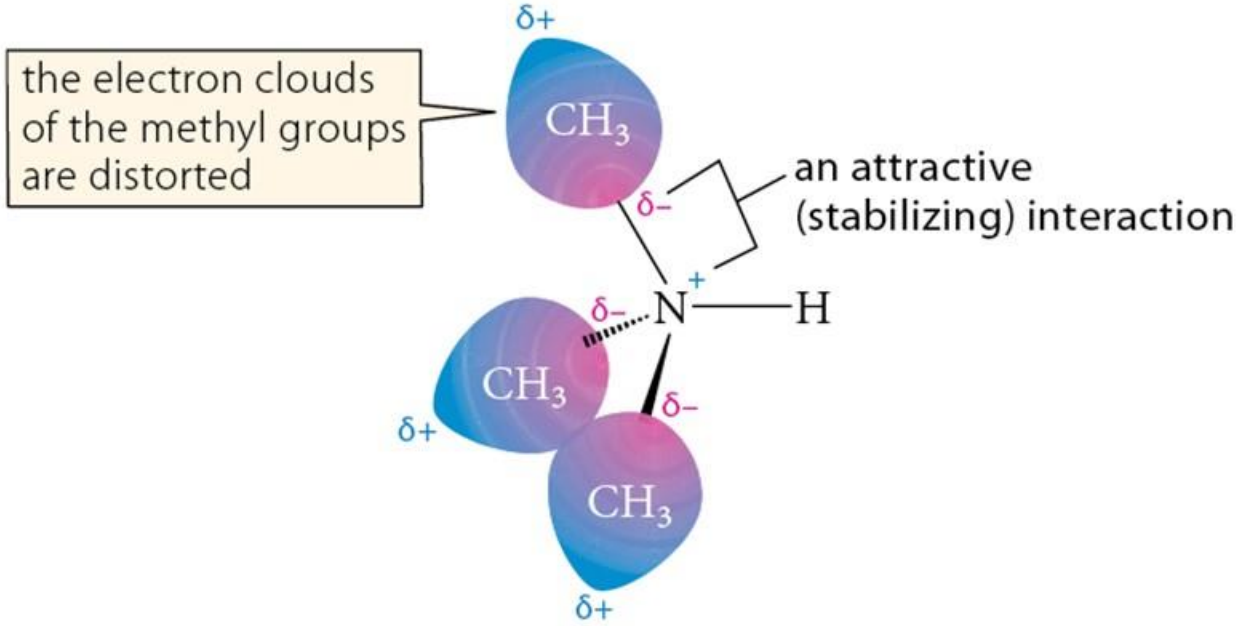
Polarization Effect of Substituents on Amine Basicity

- Alkyl substitution stabilizes charge through *polarization* effect:



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- Gas-phase acidity displays a consistent trend:

Gas-phase acidity: $\overset{+}{N}H_4 > Me\overset{+}{N}H_3 > Me_2\overset{+}{N}H_2 > Me_3\overset{+}{N}H$

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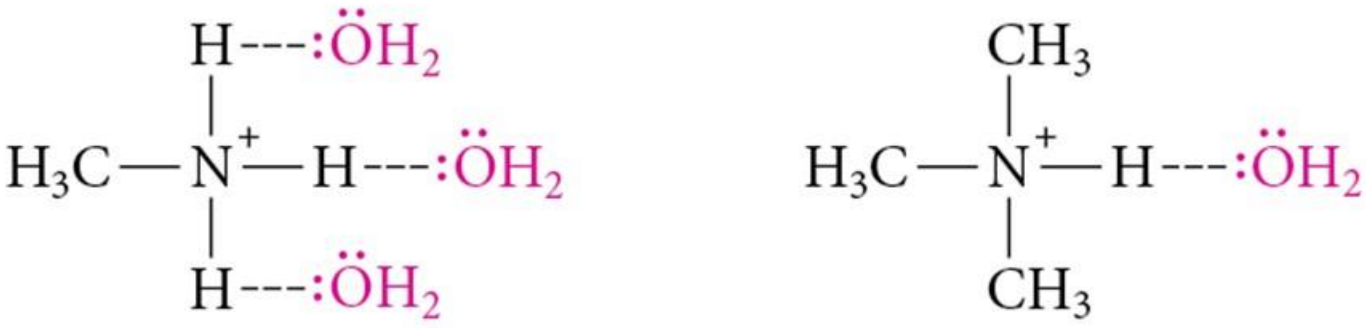
Solvent Effect on Amine Basicity

- The basicity order of amines in the aqueous solution is different from that gas phase.



- Ammonium ions in solution are stabilized not only by alkyl groups, but also by hydrogen-bond donation to the solvent.

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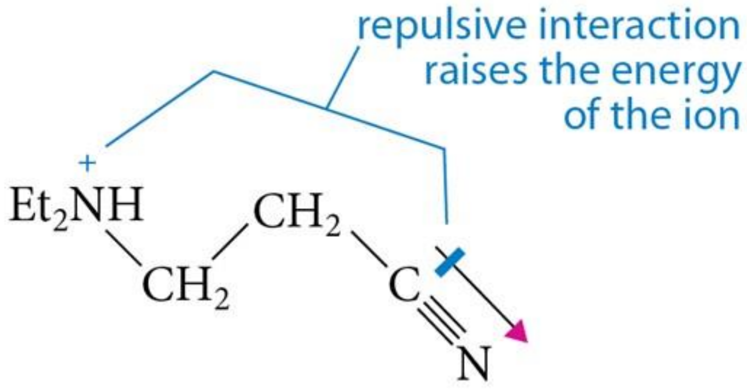
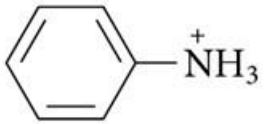
Additional Substituent Effects on Amine Basicity

- Polar effects:

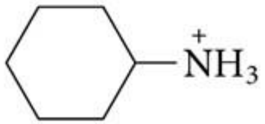
	$Et_2\overset{+}{N}HCH_2C\equiv N$	$Et_2\overset{+}{N}HCH_2CH_2C\equiv N$	$Et_2\overset{+}{N}H(CH_2)_4C\equiv N$	$Et_3\overset{+}{N}H$
pK_a:	4.55	7.65	10.08	10.65



- Resonance effects:



conjugate acid of aniline



conjugate acid of cyclohexylamine

pK_a: 4.62 (conjugate acid of aniline); 10.64 (conjugate acid of cyclohexylamine)

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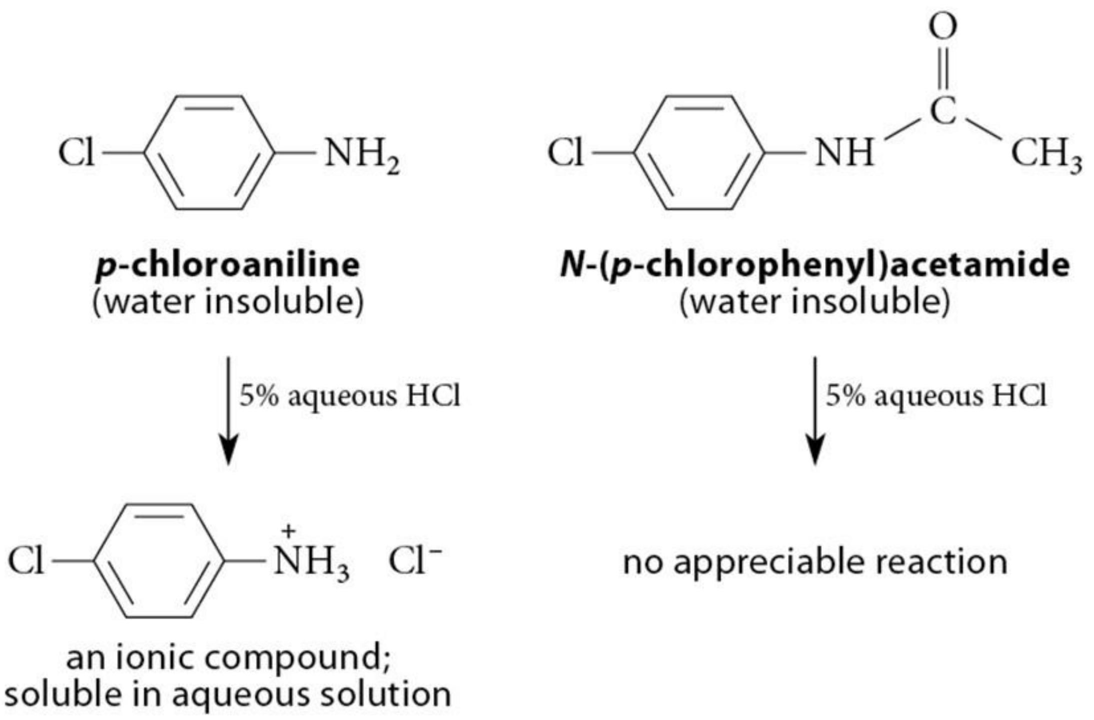
Separations Using Amine Basicity

- Ammonium salts are ionic compounds which imparts a high degree of water-solubility.
- This property can be useful in separation of amines from other compounds.



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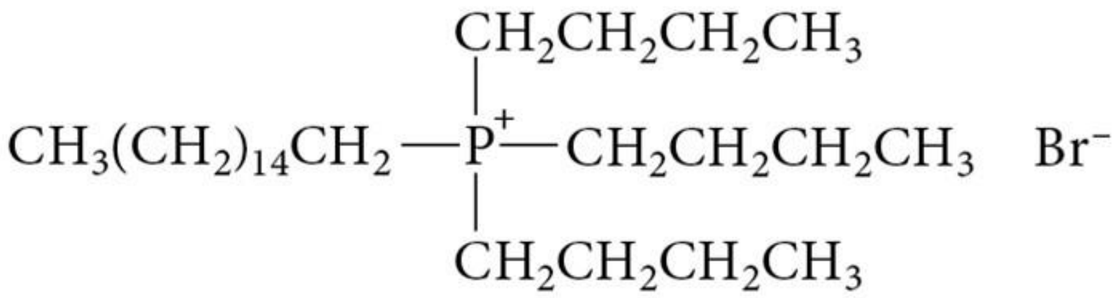
Quaternary Phosphonium Salts

- ***Quaternary phosphonium salts*** are the phosphorus analogs of quaternary ammonium salts.
- Both quaternary ammonium and phosphonium salts are used to catalyze reactions between an ionic (water-soluble) reactant and an organic (water-insoluble) reactant.
- ***Phase transfer catalysts***



cetyltributylphosphonium bromide
(a quaternary phosphonium salt)

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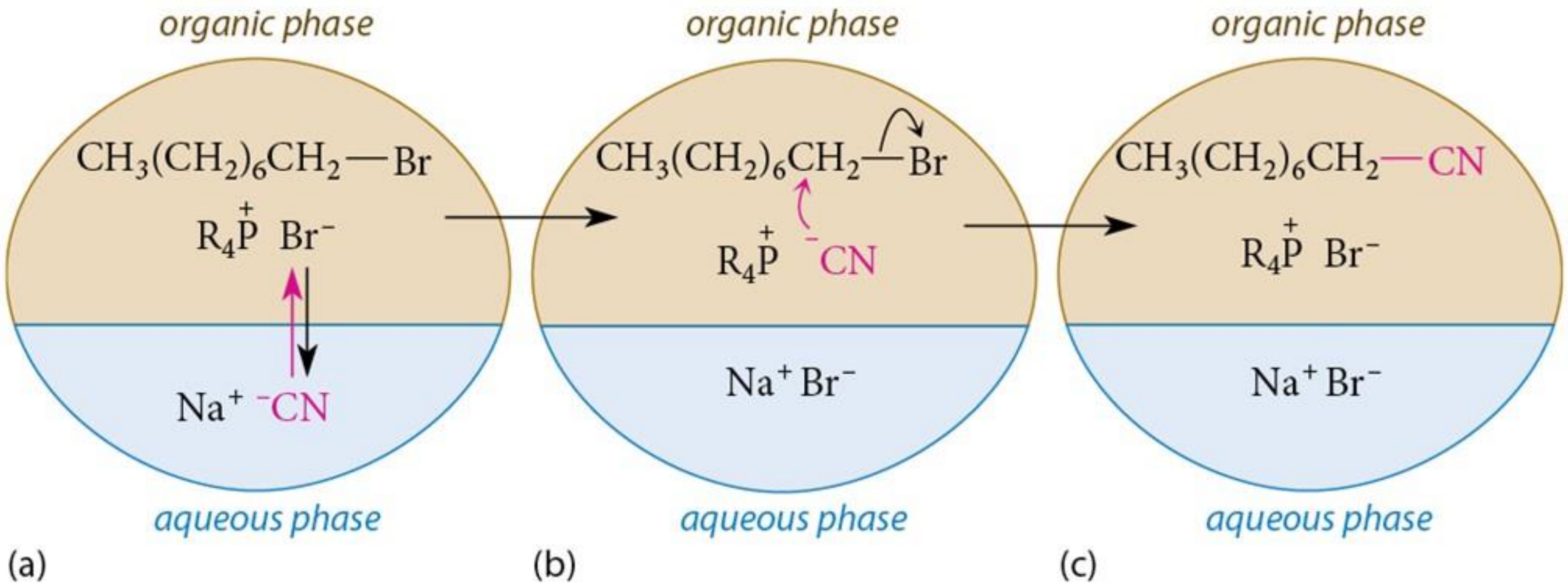
Phase-Transfer Catalysis

$$\underset{\textbf{1-bromooctane}}{CH_3(CH_2)_6CH_2{-}Br} + Na^+\ {}^-CN \xrightarrow[H_2O]{\text{quaternary phosphonium salt}} \underset{\textbf{nonanenitrile}}{CH_3(CH_2)_6CH_2{-}CN} + Na^+\,Br^-$$



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